

****Multiple Choice Questions:****

1. What is the derivative of $\sec(x)$?

- (a) $\cos(x)\tan(x)$
- (b) $\tan(x)\sec(x)$
- (c) $\cot(x)\csc(x)$
- (d) $\csc(x)\sec(x)$

2. Which of the following is the definition of the derivative of a function $f(x)$?

- (a) $\lim_{h \rightarrow 0} \frac{f(x) - f(x+h)}{h}$
- (b) $\lim_{\Delta x \rightarrow 0} \frac{f(x) - f(x+\Delta x)}{\Delta x}$
- (c) $\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$
- (d) $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h^2}$

3. What is the derivative of e^{uv} ?

- (a) e^{uv}
- (b) $e^{uv}u'v'$
- (c) $e^{uv}(u'v + uv')$
- (d) $e^{u'v'}$

4. What is the definition of $\sinh(x)$?

- (a) $\frac{e^x + e^{-x}}{2}$
- (b) $\frac{e^x - e^{-x}}{2}$
- (c) $e^x - e^{-x}$
- (d) $e^x + e^{-x}$

5. Which rule is used to find the derivative of $\frac{u}{v}$?

- (a) Chain rule
- (b) Product rule
- (c) Sum rule
- (d) Quotient rule

****Free Response Questions:****

1. Find the derivative of $f(x) = x^3 \sin(x)$ using the product rule.

2. Find the derivative of y with respect to x if $y = e^{x^2 + 2x}$.

3. Given the equation $x^2 + y^2 = 25$, use implicit differentiation to find $\frac{dy}{dx}$.

****Answer Key:****

****Multiple Choice:****

- 1. (b)
- 2. (c)
- 3. (c)
- 4. (b)
- 5. (d)

****Free Response:****

1. $f'(x) = 3x^2\sin(x) + x^3\cos(x)$

2. $\frac{dy}{dx} = e^{x^2 + 2x}(2x + 2)$

3. $\frac{dy}{dx} = -\frac{x}{y}$